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UNIVERSITY

IILM UNIVERSITY
GREATER NOIDA, UTTAR PRADESH

CODEALPHA VIRTUAL INTERNSHIP

C++ Programming Internship

Report of CGPA Calculator, Banking System & Sudoku Solver

INTERNSHIP DURATION

20 August 2026 – 20 September 2026

STUDENT CREDENTIALS

Aditya Agrawal

Roll No.

25SCS1003004654

Program

B.Tech CSE (AI & ML)

HOST ORGANIZATION

CodeAlpha

Virtual Internship Program – C++ Programming

Candidate & Program Overview

Candidate Details

Student Name: ADITYA AGRAWAL

Roll Number: 25SCS1003004654

Degree: B.Tech (Computer Science & Engineering – AI & ML)

Batch: 2025 – 2029

University: IILM University, Greater Noida, U.P.

Internship Profile

Program Name: CodeAlpha Virtual Internship – C++ Programming

Tasks Completed: 3 of 4 (CGPA Calculator, Banking System, Sudoku Solver)

Duration: 20 August 2026 – 20 September 2026

Student ID: CA/DF1/245935

Issuing Authority: Swati Srivastava, Founder & CEO, CodeAlpha

Organization Profile

CodeAlpha is a software development company offering virtual internship programs across multiple technology domains, with a focus on innovation, performance, and scalable solutions. The C++ Programming track helps interns master core language concepts, build efficient algorithms, and develop high-performance applications under mentorship.

Internship Offer Letter

Completion Certificate (QR Verified)

Unique ID Certificate

Letter of Recommendation

Job Opportunities / Placement Support

Resume Building Support

Problem Statement

Students often learn programming through isolated textbook exercises that do not reflect how software is actually designed, built, tested, and documented in practice. The internship addressed this by requiring fully working, independently testable C++ console applications solving realistic problems.

CGPA Calculator

Manually calculating GPA/CGPA across semesters is repetitive and error-prone.

Banking System

Banks need safe deposit/withdrawal handling, fund transfers, and accurate transaction history.

Sudoku Solver

Solving Sudoku by hand is slow and error-prone for harder puzzles.

Project Objectives

1

Gain hands-on C++ experience beyond coursework

2

Apply OOP principles to a real project

3

Implement a classic algorithm (backtracking)

4

Write clean, validated, modular code

5

Practice full dev cycle: design → test → document

6

Build a verifiable, working project portfolio

Technologies & Tools Used

Language

C++ (C++17 standard)

Compiler

g++ (GNU Compiler Collection)

Editor

Visual Studio Code

Version Control

Git & GitHub

Standard Library

iostream, vector, string, iomanip, ctime

Documentation

Markdown, Microsoft Word

System Architecture

CGPA Calculator

Structs, vectors, functions, input validation.
Course struct + reusable validation functions
compute quality points and CGPA.

Banking System

OOP: Customer, Account, Transaction, Bank
classes. Encapsulation & composition;
overdraft protection built in.

Sudoku Solver

2D array + recursive backtracking.
Row/column/3×3-box validation before
every placement; auto-detects invalid or
unsolvable puzzles.

Methodology & Challenges Overcome

Each project followed the same process: design data structures first, implement with input validation throughout, compile with warnings enabled, then verify with real test runs.

CHALLENGE

Duplicate transaction logging

RESOLUTION

Banking System — fixed with dedicated transfer-only logging methods

CHALLENGE

Crashes on bad input

RESOLUTION

All 3 projects — reusable validation helpers that re-prompt instead of failing

CHALLENGE

Detecting invalid Sudoku input

RESOLUTION

Sudoku Solver — pre-check added to separate “bad input” from “no solution”

Expected Outcomes & Skills Developed

Technical Skills

- C++17: structs, classes, vectors, recursion
- OOP: encapsulation, composition
- Algorithm design: recursive backtracking
- Input validation & defensive programming
- Debugging & root-cause testing
- Git & GitHub

Soft Skills & Outcomes

- Requirement analysis & specification-following
- Structured problem-solving & systematic testing
- Clear technical documentation
- Time management across deliverables
- 3 fully tested, working C++ applications
- A verifiable, GitHub-ready portfolio

Key Achievements

3

Complete C++ projects built, tested
& documented

0

Compiler warnings across all
projects (g++ -Wall)

1

Real bug found & fixed through
genuine testing

4

Weeks of internship duration
completed

Thank You!

Aditya Agrawal

Roll No: 25SCS1003004654

**School of Computer Science &
Engineering**

IILM University, Greater Noida, U.P.